

COVID-19 Vaccination Analysis

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Date

Abstract

Write your abstract here.

Keywords

Vaccine Efficacy, Bayesian Inference, Likelihood Inference, Likelihood Ratio Test

Introduction / Background

The rise of COVID-19 in 2019 resulted in a global crisis affecting healthcare, economics, and daily life across societies worldwide, forcing governments to implement unprecedented public health measures. By the end of 2020, countries such as the United States had seen over a million cases of COVID. This meant the race to find a reliable vaccine was on and it became a primary goal for pharmaceuticals and governments worldwide in order to control the pandemic. Pfizer and BioNTech developed BNT162b2, an mRNA based vaccine, while Janssen and AstraZeneca did not.

The clinical trial was a placebo-controlled, observer-blinded study that enrolled 43,548 participants aged 16 year or older. The participants were randomly assigned in a 1:1 ratio to receive either two doses of the BNT162b2 vaccine or a placebo, both administered 21 days apart. The primary endpoint was laboratory-confirmed COVID-19 occurring at least 7 days after the second dose (Polack et al., 2020).

The results from the trial are stated in the table below:

Group	Cases	Sample_size
BNT162b2	8	17,411
Placebo	162	17,511
Total	170	34,922

Table 1: Vaccine Efficacy Against COVID-19 at Least 7 Days After Second Dose

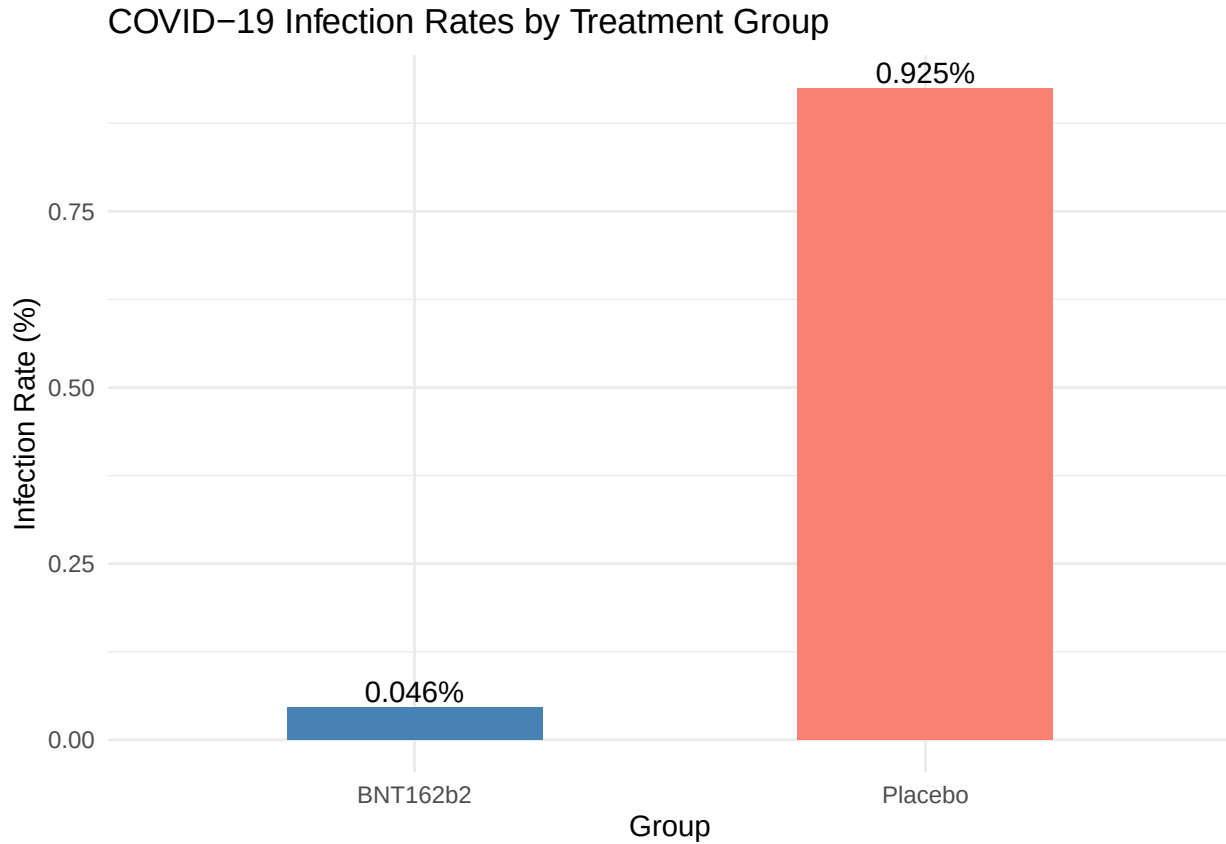
The results from the trial are stated in the table below:

Table 2: Contingency Table

Group	Cases	No_Cases	Total
BNT162b2 (Vaccine)	8	17403	17411
Placebo	162	17349	17511
Total	170	34752	34922

The total number of people is less than the one in the study because some people left the study or were not reached out after.

From the table above and the plot below, we can see that the estimated infection rates are very different. For vaccine group, we have: $\hat{\pi}_v = \frac{7}{17411} = 0.046\%$ and for the placebo group, we have $\hat{\pi}_p = \frac{162}{17511} = 0.925\%$.



As the probabilities are so small, we can look at the relative risk which can be calculated as:

$$\widehat{RR} = \frac{\pi_v}{\pi_p} = \frac{0.046\%}{0.925\%} \approx 0.0497$$

In other words, The risk of Covid 19 case in vaccine group is roughly 5% that of placebo group. Another way of saying this is that the Vaccine the risk of infection by 95%. This is called vaccine efficacy.

$$\text{vaccine efficacy} = 1 - \widehat{RR} = 1 - \frac{\pi_v}{\pi_p}$$

The US FDA requires that a new vaccine demonstrate at least 30% efficacy for approval. Polack et al. (2020) reports a vaccine efficacy of 95.0% with a 95% Bayesian credible interval of [90.3% 97.6%] and they used a beta binomial model. In this paper, we intend to extend their analysis by using both Frequentist (likelihood-based) and Bayesian methods to estimate the efficacy ψ and test the hypothesis $H_0 : \psi = 0.30$ vs $H_1 : \psi > 0.30$

Statistical Methods

Model

Let X denote the number of COVID-19 cases in the vaccine group among the $n_1 = 17,411$ vaccinated with BNT162b2 participants, and let Y denote the number of cases in the placebo group among the $n_2 = 17,511$ placebo participants. We assume

$$X \sim \text{Binom}(n_1, \pi_v) \text{ independently of } Y \sim \text{Binom}(n_2, \pi_p)$$

where π_v and π_p are the true infection probabilities in the vaccine and placebo groups, respectively.

The parameter of interest is the vaccine efficacy:

$$\psi = 1 - \frac{\pi_v}{\pi_p} = \frac{\pi_p - \pi_v}{\pi_p}$$

Rather than working with two binomial random variables, we adopt a single binomial model.

Let's look at an alternative model that we should use instead for this scenario. Let T denote the number of vaccine-group cases among the $n = 170$ total confirmed COVID-19 cases, conditional on $X + Y = 170$. The distribution of T be quite approximated by a binomial distribution.

Then

$$T \sim \text{Binom}(n = 170, \pi)$$

where $\pi = P(\text{aspirin}|\text{case})$ and can be shown to be equal to

$$\pi = \frac{n_1\pi_v}{n_1\pi_v + n_2\pi_p} = \frac{\pi_v}{\pi_v + \pi_p}$$

The vaccine efficacy ψ can be written in terms of π as:

$$\psi = \frac{1 - 2\pi}{1 - \pi}, \quad -\infty < \psi < 1$$

We can interpret ψ understanding that:

1. When $\pi = 0$ (no cases in the vaccinated group), the efficacy is perfect: $\psi = 1$.
2. When $\pi = 0.5$ (incidence is equal in both vaccinated and control groups), there is no efficacy: $\psi = 0$.
3. If $\psi < 0$, it indicates the vaccine is harmful and should definitely not be used.

From the table, we observe $t = 8$ vaccine cases among the $n = 170$ total cases. This seems very small, but we should still conduct tests to verify.

We are interested in an interval estimate for this parameter (ψ).

The hypothesis test is: $H_0 : \psi = 0.30$ versus $H_1 : \psi > 0.30$ since the FDA requires at least 30% efficacy for a new therapy to be approved.

Likelihood Inference

Detail the likelihood approach.

Bayesian Inference

In a Bayesian framework, we specify a prior distribution for π . A common choice for the binomial model is the Beta prior:

$$g(\pi) = \text{Beta}(a, b)$$

For prior elicitation, we choose a skeptical prior that reflects uncertainty about whether the vaccine is effective. This allows the data to play a larger role in determining the posterior distribution and helps prevent incorrectly concluding that the vaccine is effective when it may not be.

Coming up with two prior beliefs about ψ :

1. For $\psi \geq 0.30$, we have

$$\frac{1 - 2\pi}{1 - \pi} \geq 0.30.$$

Multiply both sides by $1 - \pi$:

$$1 - 2\pi \geq 0.30(1 - \pi).$$

Expanding gives

$$1 - 2\pi \geq 0.30 - 0.30\pi.$$

Rearranging,

$$0.70 \geq 1.70\pi,$$

so

$$\pi \leq \frac{0.70}{1.70} = \frac{7}{17}.$$

Therefore,

$$P\left(\pi \leq \frac{7}{17}\right) = 0.05.$$

2. For $\psi \geq 0$, we have

$$\frac{1 - 2\pi}{1 - \pi} \geq 0.$$

Since $1 - \pi > 0$ for $0 < \pi < 1$, this is equivalent to

$$1 - 2\pi \geq 0.$$

Hence,

$$\pi \leq \frac{1}{2}.$$

Therefore,

$$P\left(\pi \leq \frac{1}{2}\right) = 0.50.$$

So we want a $Beta(a, b)$ prior for π where the median is $\frac{1}{2}$ and the

Results

Present your findings.

Discussion / Conclusion

Discuss / conclude here.

Bibliography

Brown, B. (2024). *Lecture Title*. Lecture slides, Course Name, University Name.

Doe, J. (2020). Title of the Paper. *Journal Name*, 12(3), 45-67.

Last, F., & Last, F. (2025). *Book Title*. Publisher.

Smith, A., & Johnson, C. (2023). *Title of the Online Article*. Retrieved from <https://www.example.com>.

Appendix

Code

```
vacc_cas <- 8
plac_cas <- 162
n_vacc <- 17411
n_plac <- 17511
tot_cas <- vacc_cas + plac_cas
n_tot <- n_vacc + n_plac
```

```

table_df <- data.frame(
  Group = c("BNT162b2 (Vaccine)", "Placebo", "Total"),
  Cases = c(vacc_cas, plac_cas, tot_cas),
  No_Cases = c(n_vacc - vacc_cas, n_plac - plac_cas, n_tot - tot_cas),
  Total = c(n_vacc, n_plac, n_tot)
)

kable(table_df,
      caption = "Contingency Table",
      col.names = c("Group", "Cases", "No_Cases", "Total"),
      align = "lccc")

bar_data <- data.frame(
  Group = c("BNT162b2", "Placebo"),
  Rate = c(vacc_cas / n_vacc * 100, plac_cas / n_plac * 100)
)

ggplot(bar_data, aes(x = Group, y = Rate, fill = Group)) +
  geom_bar(stat = "identity", width = 0.5) +
  scale_fill_manual(values = c("BNT162b2" = "steelblue", "Placebo" = "salmon")) +
  labs(x = "Group", y = "Infection Rate (%)",
       title = "COVID-19 Infection Rates by Treatment Group") +
  theme_minimal() +
  theme(legend.position = "none") +
  geom_text(aes(label = paste0(round(Rate, 3), "%")), vjust = -0.25, size = 4)

```

Proofs

If applicable, include detailed mathematical derivations or additional theoretical explanations.